Week 1: BattleOps Guided Assignments

Core Infra Foundations & Cloud-Native Deep Dive

ASSIGNMENT 1: System Exploration & Debugging Basics

Goal: Understand system behavior like a DevOps engineer.

Steps

1. List all running services: **systemctl list-units --type=service**
2. Pick one service (sshd) and inspect: **systemctl status sshd || journalctl -u sshd**
3. Observe processes: **top || ps aux | grep ssh**
4. Identify: **CPU usage, Memory usage, Running services**

Outcome

You should explain:

- What is running on your system
- Which services are critical

ASSIGNMENT 2: Network + DNS Debugging

Goal: Learn how to debug connectivity issues step-by-step.

Steps

1. Test DNS: dig google.com
2. Test connectivity: ping google.com
3. Test HTTP: curl -I https://google.com
4. Break DNS manually: sudo vi /etc/resolv.conf
   Replace with: nameserver 127.0.0.2
5. Repeat tests and observe failures.

Outcome

You must explain:

- Why dig fails
- Why curl fails
- Difference between DNS and network failure

ASSIGNMENT 3: SSH + Terraform Basic Infra Setup

Goal: Set up a simple infra and connect securely.

Steps

1. Launch EC2 instance
2. Connect using SSH: ssh -i key.pem ec2-user@<IP>
3. Create a new user:
    - sudo useradd devuser
    - sudo passwd devuser

4. Add sudo: sudo usermod -aG wheel devuser
5. Run Terraform locally: terraform init, terraform apply

Outcome

You must demonstrate:

- SSH access
- User creation
- Basic infra provisioning